



Penrith Selective High School

2017

Trial Higher School Certificate
Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A Reference Sheet is provided
- In Questions 11-14, show relevant mathematical reasoning and/or calculations in the writing booklets provided
- All diagrams are not to scale

Total Marks – 70

Section I Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Assessor: X. Chirgwin

Student Number: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2017 Higher School Certificate Examination.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

Q1. A is the point $(2, -3)$ and B is the point $(-5, 11)$. Which of the following are the coordinates of a point that divides AB internally in the ratio of 4:3?

(A) $(3, -1)$

(B) $(-1, 3)$

(C) $(5, -2)$

(D) $(-2, 5)$

Q2. What is the natural domain of the function $f(x) = 2\sqrt{x+1} - \sqrt{x-2}$?

(A) $x \leq -1$

(B) $x \geq 2$

(C) $x \leq -1$ or $x \geq 2$

(D) $-1 \leq x \leq 2$

Q3. What is the acute angle between the tangents drawn to the curve $y = e^{3x}$ at the points where $x = 0$ and $x = 1$ to the nearest degree?

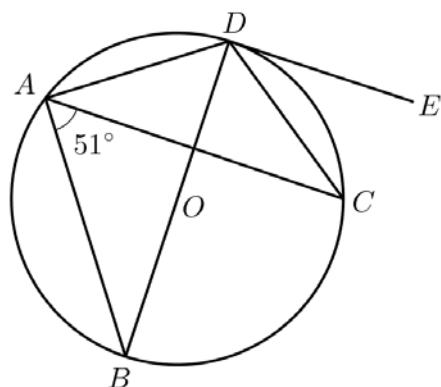
(A) 17°

(B) 19°

(C) 42°

(D) 44°

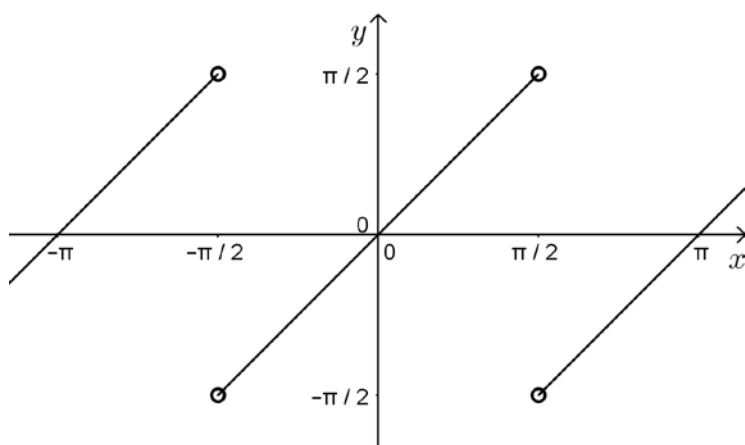
Q4.



DE is a tangent to the circle with centre O . Given that $\angle BAC = 51^\circ$, the size of $\angle EDC$ is:

- (A) 28°
- (B) 39°
- (C) 40°
- (D) 51°

Q5.



The possible equation of the graph shown above is:

- (A) $y = \tan^{-1}(\tan x)$
- (B) $y = \tan(\tan^{-1} x)$
- (C) $y = \sin(\sin^{-1} x)$
- (D) $y = \sin^{-1}(\sin x)$

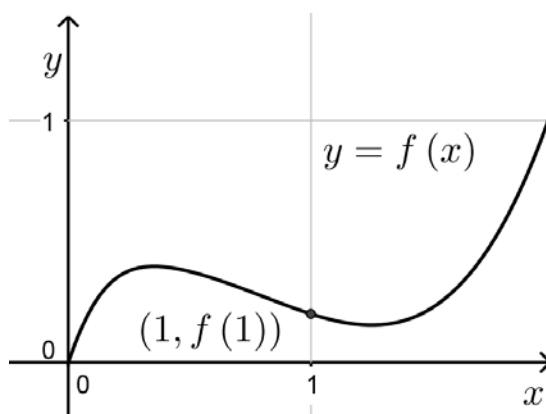
Q6. A parabola has the parametric equations $x = -8t^2$, $y = 16t$. The coordinates of the focus for this parabola is:

- (A) $(-8, 0)$
- (B) $(8, 0)$
- (C) $(0, -8)$
- (D) $(0, 8)$

Q7. The lock for a gate opens with a five-digit code. Each wheel rotates through the digits 0 to 9. The percentage of five-digit codes that have no repeated digits is closest to:

- (A) 17%
- (B) 30%
- (C) 50%
- (D) 63%

Q8. The diagram shows $y = f(x)$.



Which of the following is a correct statement?

- (A) $f(1) < f'(1) < f''(1) < 1$
- (B) $1 < f(1) < f'(1) < f''(1)$
- (C) $f''(1) < f(1) < 1 < f'(1)$
- (D) $f'(1) < f(1) < f''(1) < 1$

Q9. The polynomial $P(x)$ has degree 6 and the polynomial $Q(x)$ has degree 3. If you divide $P(x)$ by $Q(x)$, the remainder may have degree:

- (A) 0
- (B) 0 or 1
- (C) 0, 1 or 2
- (D) 0, 1, 2 or 3

Q10. An object moves in a straight line so that at time t its displacement from a fixed origin is x and its velocity is v . The acceleration is $7 - 3x^2$. Which of the following is the correct equation for the velocity given that $v = 4$ when $x = 2$?

- (A) $v = \sqrt{14x - 2x^3 + 4}$
- (B) $v = \sqrt{14x - x^3 - 4}$
- (C) $v = \sqrt{7x - x^3 + 10}$
- (D) $v = 7x - x^3 - 2$

Section II

60 Marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

a) Solve $\frac{1}{3x+2} \geq 2$ 2

b) Using the substitution $u = \cos x$, evaluate 3

$$\int_0^{\frac{\pi}{2}} 3 \sin x \cos^5 x \, dx$$

c) Two school captains, two vice captains and three other students are sitting around a circular table. What is the probability that the two school captains sit next to each other? 2

d) Find the constant term in the expansion of $\left(5x^4 - \frac{1}{2x}\right)^{10}$ 3

e) Show that $\frac{d}{dx}(xe^{\tan^{-1} x}) = \left(\frac{x^2 + x + 1}{x^2 + 1}\right)e^{\tan^{-1} x}$ 2

f) Given that $\ln|2x - 1| = \tan x$ has a root close to 4.2, use one application of Newton's method to obtain a better approximation of the root. Round your answer to 3 decimal places. 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

a) If α, β and γ are the roots of $3x^3 - 4x + 6 = 0$. Evaluate:

i) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ 2

ii) $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$ 2

b) N is the number of songbirds in a certain population at time t years. The population size N satisfies the equation $\frac{dN}{dt} = k(N - 750)$, for some constant k .

i) Verify by differentiation that $N = 750 + Ae^{kt}$ is a solution of the equation $\frac{dN}{dt} = k(N - 750)$, where A is a constant. 1

ii) Initially there are 3500 songbirds but after 3 years there are only 2400 left. Find the exact values of A and k . 2

iii) Find the population after 7 years, round your answer to the nearest whole number. 1

iv) If this trend continues, what would the value of the population eventually become? 1

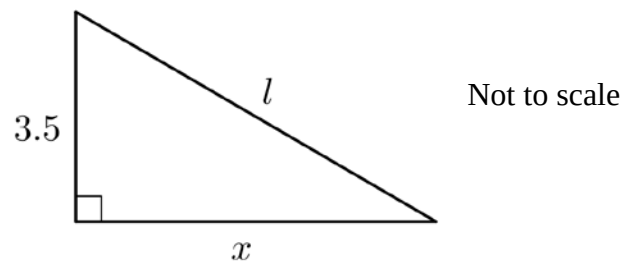
v) Hence, sketch the graph of population size against time. 1

Question 12 continues on page 8

Question 12 (continued)

- c) A boat is attached by a rope to a jetty 3.5 metres above the front end of the boat. The boat is being pulled in towards the jetty by the rope at a rate of 1 m/s.

Let x be the horizontal distance between the boat and the jetty, and l be the length of the rope as shown below.



- i) Show that $2l = \sqrt{49 + 4x^2}$ 1
- ii) Show that the change in the position of the boat is given by 2
- $$\frac{dx}{dt} = -\frac{\sqrt{49 + 4x^2}}{2x}$$
- iii) At what rate is the boat approaching the jetty when 4 metres of rope still remains to be pulled in? Leave your answer in exact form. 2

End of Question 12

Question 13

(15 marks) Use a SEPARATE writing booklet.

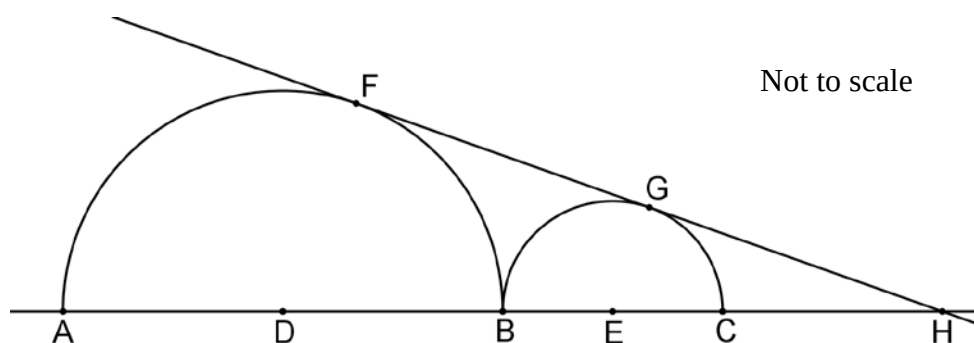
- a) Prove by mathematical induction that

3

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

n is any positive integer, where $n \geq 1$.

- b) The diagram below shows two semicircles centred at D and E with radii 5 cm and 3 cm respectively. FGH is a common tangent to both semicircles.



- i) Prove that $\triangle GEH \parallel \triangle FDH$. **2**
- ii) Show that EH is 12 cm. **1**
- iii) Show that FBG lie on another circle with FG as its diameter. **3**

Question 13 continues on page 11

Question 13 (continued)

- c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are extremities of a focal chord for the parabola $x^2 = 4ay$.
- i) Given that the equation of the chord PQ is $2y = (p + q)x - 2apq$ [DO NOT PROVE THIS], show that $pq = -1$. 1
- ii) Given that the equation of the tangent at P is $y = px - ap^2$ and the tangent at Q is $y = qx - aq^2$ [DO NOT PROVE THESE]. 2
- Show that the tangents at P and Q meet at $R(a(p + q), -a)$.
- iii) Hence state the locus of R . 1
- iv) Show that the chord PQ has length 2
- $$a\left(p + \frac{1}{p}\right)^2$$

End of Question 13

Question 14

(15 marks) Use a SEPARATE writing booklet.

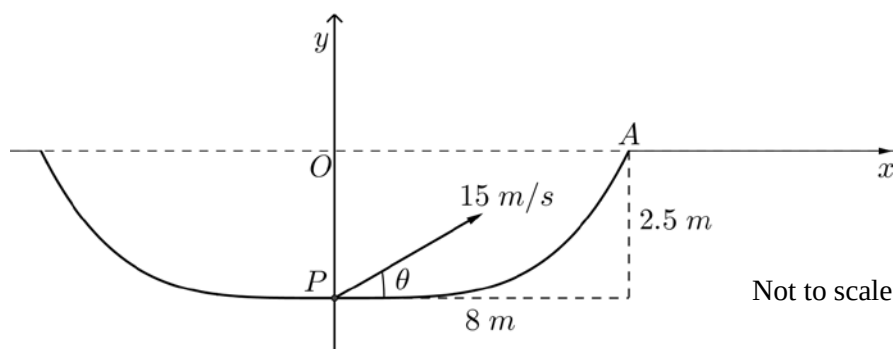
a) Find the exact value of $\sin\left[\cos^{-1}\left(\frac{3}{4}\right) - \tan^{-1}\left(-\frac{2}{7}\right)\right]$ 3

b) Given that $(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$, show that:

i) $\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$ 1

ii) $\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{2^{2n+1} - 1}{2n+1}$ 3

- c) A golf ball is lying at P , at the middle of the bottom of a sand bunker which is surrounded by level ground. The point A is at the edge of the bunker 8 m from O . The golf ball is hit with initial velocity of 15 m/s and P is 2.5 m below O . Ignoring air resistance.



- i) Using $g = -10\text{ m/s}^2$, show that the golf ball's trajectory at time t seconds after being hit may be defined by the equations: 2

$$x = 15t \cos \theta \text{ and } y = -5t^2 + 15t \sin \theta - 2.5$$

where x and y are the horizontal and vertical displacements, in metres, of the ball from the origin O shown in the diagram, and θ is the angle of projection.

- ii) Given $\theta = 30^\circ$, how far to the right of A will the ball land? Round your answer to 2 decimal places. 3
- iii) Find the range of values of θ , to the nearest degree, at which the ball must be hit so that it will land to the right of A . 3

End of Paper

2017 Mathematics Extension 1 Trial Solution

Section 1:

Q1. D Q2. B Q3. A Q4. B Q5. A Q6. A Q7. B Q8. D Q9. C Q10. A

Q1. D

$$x = \frac{4 \times (-5) + 3 \times 2}{4 + 3}$$

$$x = \frac{-14}{7}$$

$$x = -2$$

$$y = \frac{4 \times 11 + 3 \times (-3)}{4 + 3}$$

$$y = \frac{35}{7}$$

$$y = 5$$

$$(-2, 5)$$

Q2. B

$$f(x) = 2\sqrt{x+1} - \sqrt{x-2}$$

$$\text{For } 2\sqrt{x+1},$$

$$x+1 \geq 0$$

$$x \geq -1$$

$$\text{For } \sqrt{x-2},$$

$$x-2 \geq 0$$

$$x \geq 2$$

$$\text{For } 2\sqrt{x+1} - \sqrt{x-2}, \text{ domain must satisfy both } 2\sqrt{x+1} \text{ and } \sqrt{x-2}$$

$$\therefore x \geq 2$$

Q3. A

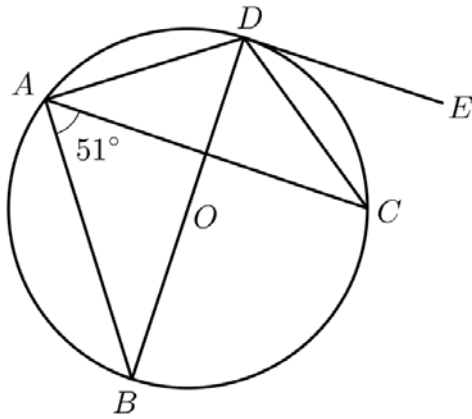
$$y = e^{3x}, y' = 3e^{3x}$$

$$\text{At } x = 0, m_1 = 3; \text{ At } x = 1, m_2 = 3e^3$$

$$\tan \theta = \frac{3e^3 - 3}{1 + 9e^3}$$

$$\theta = 17^\circ$$

Q4. B



$$\angle BDC = \angle BAC = 51^\circ \text{ (angles in the same segment)}$$

$$\angle ODE = 90^\circ \text{ (radius } OD \perp \text{ tangent } DE)$$

$$\angle EDC = \angle ODE - \angle BDC$$

$$\angle EDC = 90^\circ - 51^\circ$$

$$\angle EDC = 39^\circ$$

Q5. A

Q6. A

$$x = -8t^2, y = 16t$$

$$y^2 = -32x$$

$$a = 8$$

Focus is $(-8, 0)$

Q7. B

$$\frac{10 \times 9 \times 8 \times 7 \times 6}{10^5} \times 100\% = 30\%$$

Q8. D

$$0 < f(1) < 1, f'(1) < 0, f''(x) > 0$$

$$\therefore f'(1) < f(1) < f''(1) < 1$$

Q9. C

Q10. A

$$\ddot{x} = 7 - 3x^2$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 7 - 3x^2$$

$$\frac{1}{2} v^2 = 7x - x^3 + C$$

$$v^2 = 14x - 2x^3 + C$$

$$\text{At } x = 2, v = 4$$

$$16 = 14 \times 2 - 2 \times 2^3 + C$$

$$C = 4$$

$$v^2 = 14x - 2x^3 + 4$$

$$v = \pm \sqrt{14x - 2x^3 + 4}$$

$$\text{Given condition that } x = 2, v = 4$$

$$v > 0$$

$$\therefore v = \sqrt{14x - 2x^3 + 4}$$

Section 2

Q11.

a)

$$\frac{1}{3x+2} \geq 2 \quad x \neq -\frac{2}{3}$$

$$(3x+2) \geq 2(3x+2)^2$$

$$2(3x+2)^2 - (3x+2) \leq 0$$

$$(3x+2)(6x+4-1) \leq 0$$

$$(3x+2)(6x+3) \leq 0$$

$$\therefore -\frac{2}{3} < x \leq -\frac{1}{2}$$

b)

$$I = \int_0^{\frac{\pi}{2}} 3 \sin x \cos^5 x \, dx$$

$$\text{Let } u = \cos x$$

$$du = -\sin x \, dx$$

$$x = \frac{\pi}{2}, \quad u = 0$$

$$x = 0, \quad u = 1$$

$$I = \int_1^0 -3u^5 \, du$$

$$= \int_0^1 3u^5 \, du$$

$$= \left[\frac{3u^6}{6} \right]_0^1$$

$$= \frac{1}{2}$$

c)

7 people sitting around a table is $6!$

Two school captains sitting next to each other is $2! \times 5!$

$$P = \frac{2! \times 5!}{6!} = \frac{1}{3}$$

d)

$$\begin{aligned} {}^{10}C_r (5x^4)^{10-r} \times \left(-\frac{1}{2}x^{-1}\right)^r &= {}^{10}C_r 5^{10-r} \times \left(-\frac{1}{2}\right)^r \times x^{40-4r} \times x^{-r} \\ &= {}^{10}C_r 5^{10-r} \times \left(-\frac{1}{2}\right)^r \times x^{40-5r} \end{aligned}$$

For the constant term,

$$\begin{aligned} 40 - 5r &= 0 \\ r &= 8 \end{aligned}$$

$${}^{10}C_8 5^{10-8} \times \left(-\frac{1}{2}\right)^8 = \frac{1125}{256}$$

e)

$$\begin{aligned} \frac{d}{dx}(xe^{\tan^{-1}x}) &= \frac{xe^{\tan^{-1}x}}{x^2+1} + e^{\tan^{-1}x} \\ &= \left(\frac{x}{x^2+1} + 1\right)e^{\tan^{-1}x} \\ &= \left(\frac{x^2+x+1}{x^2+1}\right)e^{\tan^{-1}x} \end{aligned}$$

f)

$$\ln|2x-1| = \tan x$$

$$f'(x) = \frac{2}{2x-1} - \sec^2 x$$

$$x_1 = 4.2 - \frac{\ln(2 \times 4.2 - 1) - \tan 4.2}{\frac{2}{2 \times 4.2 - 1} - \sec^2 4.2}$$

$$x_1 = 4.258 \text{ (3 d. p.)}$$

Question 12

a) i)

$$3x^3 - 4x + 6 = 0$$

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -\frac{4}{3}$$

$$\alpha\beta\gamma = -\frac{6}{3} = -2$$

$$\begin{aligned}\alpha^{-1} + \beta^{-1} + \gamma^{-1} &= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \\ &= \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma} \\ &= \frac{-\frac{4}{3}}{-2} \\ &= \frac{2}{3}\end{aligned}$$

ii)

$$\begin{aligned}\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3 &= \alpha\beta\gamma(\alpha^2 + \beta^2 + \gamma^2) \\ &= \alpha\beta\gamma[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)] \\ &= -2\left[0 - 2 \times \left(-\frac{4}{3}\right)\right] \\ &= -\frac{16}{3}\end{aligned}$$

b) i)

$$\begin{aligned}N &= 750 + Ae^{kt} \\ \frac{dN}{dt} &= kAe^{kt} \\ \frac{dN}{dt} &= k(N - 750)\end{aligned}$$

ii)

$$\text{When } t = 0, N = 3500$$

$$\begin{aligned}3500 &= 750 + Ae^0 \\ \therefore A &= 2750\end{aligned}$$

$$\text{When } t = 3, N = 2400$$

$$\begin{aligned}2400 &= 750 + 2750e^{3k} \\ 1650 &= 2750e^{3k} \\ \frac{1650}{2750} &= e^{3k} \\ \ln\left(\frac{3}{5}\right) &= 3k \\ k &= \frac{\ln 0.6}{3}\end{aligned}$$

iii)

$$N = 750 + 2750e^{7 \times \frac{\ln 0.6}{3}}$$

$$N \approx 1585$$

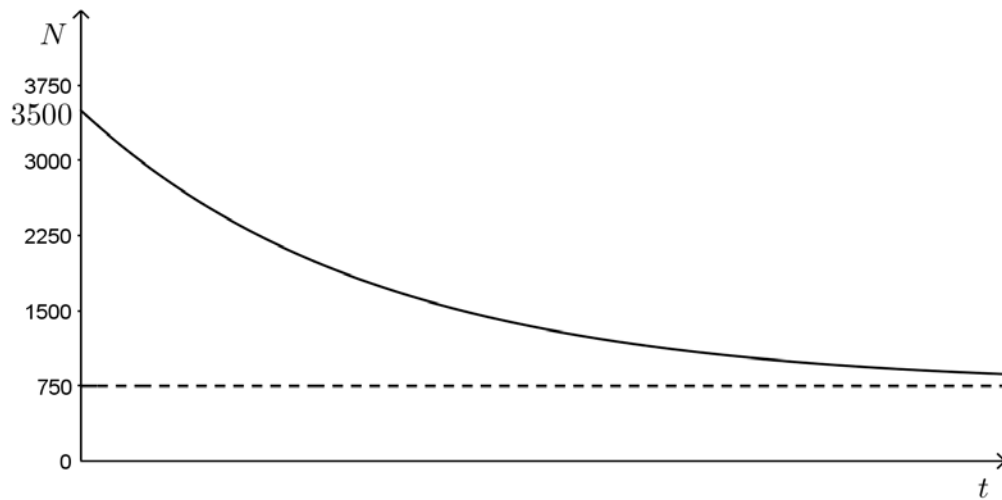
iv)

Since $k = -0.170 \dots$

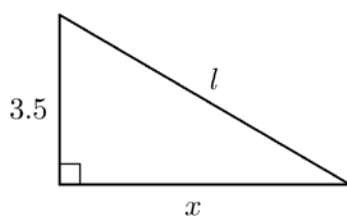
As $t \rightarrow +\infty$, $e^{kt} \rightarrow 0$

$$\therefore N \rightarrow 750$$

v)



c) i)



$$l^2 = 3.5^2 + x^2 \text{ (by Pythagoras' theorem)}$$

$$l = \sqrt{\frac{49 + 4x^2}{4}}$$

$$l = \frac{1}{2}\sqrt{49 + 4x^2}$$

$$2l = \sqrt{49 + 4x^2}$$

ii)

$$\frac{dl}{dx} = \frac{1}{2} \times \frac{1}{2} \times 8x \times \frac{1}{\sqrt{49 + 4x^2}}$$

$$\frac{dl}{dx} = \frac{2x}{\sqrt{49 + 4x^2}}$$

$$\frac{dl}{dt} = \frac{dl}{dx} \times \frac{dx}{dt}$$

$$-1 = \frac{2x}{\sqrt{49 + 4x^2}} \times \frac{dx}{dt}$$

$$\frac{dx}{dt} = -\frac{\sqrt{49 + 4x^2}}{2x}$$

iii)

At $l = 4$

$$16 = 3.5^2 + x^2$$

$$x = \sqrt{\frac{15}{4}}$$

$$x = \frac{\sqrt{15}}{2}$$

$$\frac{dx}{dt} = -\frac{\sqrt{49 + 4 \times \frac{15}{4}}}{2 \times \frac{\sqrt{15}}{2}}$$

$$\frac{dx}{dt} = -\frac{\sqrt{49 + 4 \times \frac{15}{4}}}{2 \times \frac{\sqrt{15}}{2}}$$

$$\frac{dx}{dt} = -\frac{8}{\sqrt{15}} \text{ m/s}$$

The boat is approaching the jetty at a rate of $\frac{8}{\sqrt{15}}$ m/s

Q13

a)

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

1. Prove statement is true for $n = 1$

$$LHS = \frac{1}{2!} = \frac{1}{2}$$

$$RHS = \frac{(1+1)! - 1}{(1+1)!} = \frac{1}{2}$$

$$LHS = RHS$$

 $\therefore$ Statement is true for $n = 1$ 2. Assume statement is true for $n = k$ where k is some positive integer

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!}$$

3. Prove statement is true for $n = k + 1$

$$i.e. \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = \frac{(k+2)! - 1}{(k+2)!}$$

$$LHS = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$LHS = \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{from step 2})$$

$$LHS = \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$LHS = \frac{(k+1)! - 1}{(k+1)!} \times \frac{(k+2)}{(k+2)} + \frac{k+1}{(k+2)!}$$

$$LHS = \frac{(k+2)! - (k+2)}{(k+2)!} + \frac{k+1}{(k+2)!}$$

$$LHS = \frac{(k+2)! - (k+2) + k+1}{(k+2)!}$$

$$LHS = \frac{(k+2)! - k - 2 + k + 1}{(k+2)!}$$

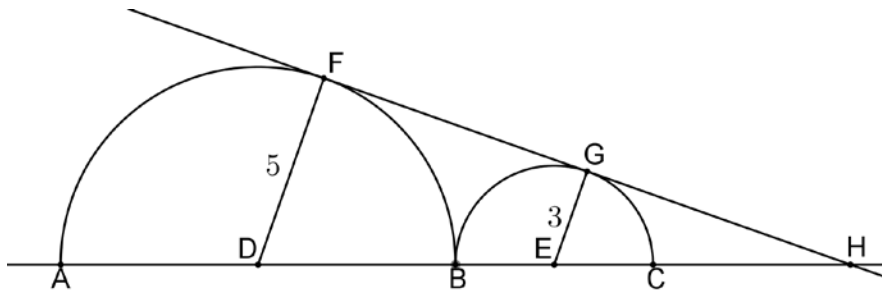
$$LHS = \frac{(k+2)! - 1}{(k+2)!}$$

$$LHS = RHS$$

 $\therefore$ This statement is true by mathematical induction for all positive integers n .

b)

i)



Join FD and GE

In $\triangle GEH$ and $\triangle FDH$

$\angle H$ is common

$\angle DFH = 90^\circ$ ($DF \perp FH$, tangent perpendicular to radius)

$\angle EGH = 90^\circ$ ($EG \perp FH$, tangent perpendicular to radius)

$\angle DFH = \angle EGH$

$\therefore \triangle GEH \sim \triangle FDH$ (equiangular)

ii)

$\frac{HE}{HD} = \frac{GE}{FD}$ (corresponding sides of similar triangles are in proportion)

$$\frac{HE}{HE + DB + BE} = \frac{3}{5}$$

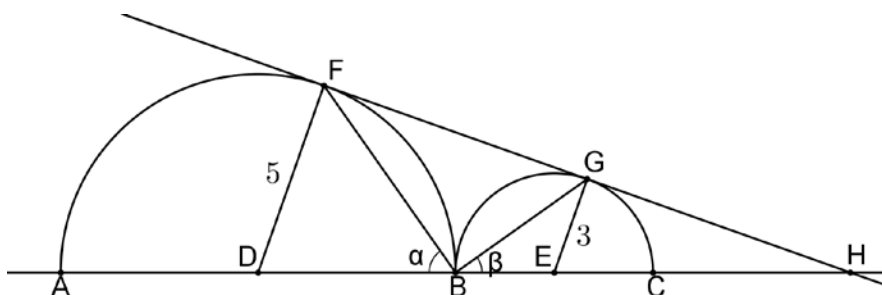
$$\frac{HE}{HE + 8} = \frac{3}{5}$$

$$5HE = 3(HE + 8)$$

$$2HE = 24$$

$$HE = 12 \text{ cm}$$

iii)



Join FB and GB

Let $\angle FBD = \alpha$ and $\angle GBE = \beta$

$DF = DB$ (equal radii)

$\triangle DBF$ is an isosceles triangle (two equal sides)

$\angle FBD = \angle DFB = \alpha$ (equal base angles of isosceles $\triangle DBF$)

$\angle DFB + \angle BFG = \angle DFH$ (complementary angles)

$\angle BFG = 90^\circ - \alpha$

$BE = GE$ (equal radii)

$\triangle BEG$ is an isosceles triangle (two equal sides)

$\angle EBG = \angle EGB = \beta$ (equal base angles of isosceles $\triangle BEG$)

$\angle EGB + \angle BGF = \angle EGF$ (complementary angles)

$\angle BGF = 90^\circ - \beta$

$\angle BGF + \angle BFG + \angle FBG = 180^\circ$ (angle sum of $\triangle FBG$)

$\angle FBG = 180^\circ - (90^\circ - \alpha) - (90^\circ - \beta)$

$\angle FBG = \alpha + \beta$

$\angle FBG + \angle FBD + \angle GBE = 180^\circ$ (angle sum on a straight line)

$\alpha + \beta + \alpha + \beta = 180^\circ$

$2(\alpha + \beta) = 180^\circ$

$\alpha + \beta = 90^\circ$

$\angle FBG = 90^\circ$

Since angle in a semicircle is 90°

$\therefore FBG$ lie on another circle with FG as its diameter.

c) i)

Since PQ is a focal chord, it must pass through the focus $(0, a)$

$$(p + q) \times 0 - 2a - 2apq = 0$$

$$-2apq = 2a$$

$$\therefore pq = -1$$

ii)

$$\text{Tangent at } P \quad y = px - ap^2 \quad (1)$$

$$\text{Tangent at } Q \quad y = qx - aq^2 \quad (2)$$

Solve (1) and (2) simultaneously to find the point of intersection.

$$px - ap^2 = qx - aq^2$$

$$x = a(p + q)$$

Substitute $x = a(p + q)$ into (1)

$$y = p \times a(p + q) - ap^2$$

$$y = ap^2 + apq - ap^2$$

$$y = apq$$

$$y = -a \quad (pq = -1)$$

$$\therefore (a(p + q), -a)$$

iii)

The locus of R is a horizontal line with equation $y = -a$.

iv)

$$PQ = \sqrt{(2ap - 2aq)^2 + (ap^2 - aq^2)^2}$$

$$PQ = \sqrt{4a^2(p - q)^2 + a^2(p^2 - q^2)^2}$$

$$PQ = a\sqrt{4(p - q)^2 + (p^2 - q^2)^2}$$

$$PQ = a\sqrt{4(p - q)^2 + (p^2 - q^2)(p^2 - q^2)}$$

$$PQ = a\sqrt{4(p - q)^2 + (p + q)^2(p - q)^2}$$

$$PQ = a(p - q)\sqrt{4 + (p + q)^2}$$

$$PQ = a(p - q)\sqrt{-4pq + p^2 + 2pq + q^2} \quad (pq = -1)$$

$$PQ = a(p - q)\sqrt{p^2 - 2pq + q^2}$$

$$PQ = a(p - q)\sqrt{(p - q)^2}$$

$$PQ = a(p - q)^2$$

$$\therefore PQ = a\left(p + \frac{1}{p}\right)^2 \quad \left(q = -\frac{1}{p}\right)$$

Q14.

a)

$$\sin \left[\cos^{-1} \left(\frac{3}{4} \right) - \tan^{-1} \left(-\frac{2}{7} \right) \right]$$

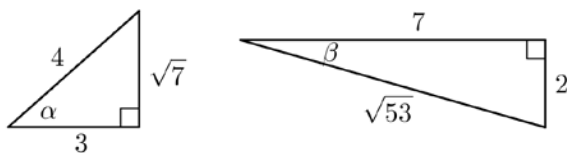
$$\text{Let } \alpha = \cos^{-1} \left(\frac{3}{4} \right), 0 \leq \alpha \leq \pi$$

$$\cos \alpha = \frac{3}{4}$$

$$\text{Let } \beta = \tan^{-1} \left(-\frac{2}{7} \right), -\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2}$$

$$\tan \beta = -\frac{2}{7}$$

α represents an angle in the first quadrant and β represents an angle in the fourth quadrant.



$$\sin \left[\cos^{-1} \left(\frac{3}{4} \right) - \tan^{-1} \left(-\frac{2}{7} \right) \right]$$

$$= \sin[\alpha - \beta]$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \frac{\sqrt{7}}{4} \times \frac{7}{\sqrt{53}} - \frac{3}{4} \times \frac{-2}{\sqrt{53}}$$

$$= \frac{7\sqrt{7}}{4\sqrt{53}} + \frac{6}{4\sqrt{53}}$$

$$= \frac{7\sqrt{371} + 6\sqrt{53}}{212}$$

b) i)

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

Substitute $x = 1$

$$\sum_{k=0}^{2n} \binom{2n}{k} 1^k = (1+1)^{2n}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} = 2^{2n}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$$

ii)

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

Integrate both sides of:

$$\frac{(1+x)^{2n+1}}{2n+1} + C = \sum_{k=0}^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1}$$

Substitute $x = 0$

$$\frac{(1+0)^{2n+1}}{2n+1} + C = \sum_{k=0}^{2n} \binom{2n}{k} \frac{0^{k+1}}{k+1}$$

$$\frac{1}{2n+1} + C = 0$$

$$C = -\frac{1}{2n+1}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1} = \frac{(1+x)^{2n+1}}{2n+1} - \frac{1}{2n+1}$$

Substitute $x = 1$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1^{k+1}}{k+1} = \frac{(1+1)^{2n+1}}{2n+1} - \frac{1}{2n+1}$$

$$\begin{aligned} \sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} &= \frac{2^{2n+1}}{2n+1} - \frac{1}{2n+1} \\ &= \frac{2^{2n+1} - 1}{2n+1} \end{aligned}$$

c) i)

Horizontal:

$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

$$\text{When } t = 0, \dot{x} = 15 \cos \theta, \therefore C_1 = 15 \cos \theta$$

$$\dot{x} = 15 \cos \theta$$

$$x = 15t \cos \theta + C_3$$

$$\text{When } t = 0, x = 0, \therefore C_3 = 0$$

$$x = 15t \cos \theta$$

Vertical:

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_2$$

$$\text{When } t = 0, \dot{y} = 15 \sin \theta, \therefore C_2 = 15 \sin \theta$$

$$\dot{y} = -10t + 15 \sin \theta$$

$$y = -5t^2 + 15t \sin \theta + C_4$$

$$\text{When } t = 0, y = -2.5, \therefore C_4 = -2.5$$

$$y = -5t^2 + 15t \sin \theta - 2.5$$

ii)

Given that $\theta = 30^\circ$, ball will hit ground when $y = 0$

$$y = -5t^2 + 15t \sin \theta - 2.5$$

$$0 = -5t^2 + 15t \sin 30^\circ - 2.5$$

$$0 = -5t^2 + 15t \times \frac{1}{2} - 2.5$$

$$0 = -10t^2 + 15t - 5$$

$$0 = 10t^2 - 15t + 5$$

$$0 = 10t^2 - 10t - 5t + 5$$

$$0 = 10t(t - 1) - 5(t - 1)$$

$$0 = (10t - 5)(t - 1)$$

$$t = \frac{1}{2}, \quad t = 1$$

$t = \frac{1}{2}$ gives first time ball crosses x axis which is not on the ground (it is left of A)

$t = 1$ gives the time the ball hits the ground to the right of A .

When $t = 1$,

$$x = 15 \times 1 \times \cos 30^\circ$$

$$x = \frac{15\sqrt{3}}{2}$$

$$OA = 8$$

$$\frac{15\sqrt{3}}{2} - 8 \approx 4.99$$

$\therefore$ The ball will land 4.99 metres to the right of A

iii)

For the ball to land to the right of A , we need the angle necessary to go through A .

$$A(8, 0)$$

$$8 = 15t \cos \theta$$

$$t = \frac{8}{15 \cos \theta}$$

Substitute t into $y = -5t^2 + 15t \sin \theta - 2.5$

$$y = -5 \left(\frac{8}{15 \cos \theta} \right)^2 + 15 \left(\frac{8}{15 \cos \theta} \right) \sin \theta - 2.5$$

$$y = -\frac{64}{45 \cos^2 \theta} + \frac{8 \sin \theta}{\cos \theta} - 2.5$$

When $y = 0$

$$0 = -\frac{64}{45 \cos^2 \theta} + \frac{8 \sin \theta}{\cos \theta} - 2.5$$

$$0 = -128 \sec^2 \theta + 720 \tan \theta - 225$$

$$0 = 128(1 + \tan^2 \theta) - 720 \tan \theta + 225$$

$$0 = 128 \tan^2 \theta - 720 \tan \theta + 353$$

$$\tan \theta = \frac{720 \pm \sqrt{(-720)^2 - 4 \times 128 \times 353}}{2 \times 128}$$

$$\tan \theta = \frac{720 \pm 16\sqrt{1319}}{256}$$

$$\tan \theta = \frac{45 \pm \sqrt{1319}}{16}$$

$$\theta = 28^{\circ}29' \quad \text{or} \quad 78^{\circ}52'$$

Anything less than $28^{\circ}29'$ or greater than $78^{\circ}52'$ will hit the bank of the sand bunker. So to land to the right of A , $29^{\circ} \leq \theta \leq 78^{\circ}$.